

Spectral concentration in two large Vina docking-score matrices: two-way centering reveals correlated residual variation without a resolved target-ranking gain

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Abstract

Reverse docking, off-target profiling and multiobjective design use docking scores as a ligand-by-target matrix, yet shared score tendencies can obscure target-resolved variation. We measured its target-correlation spectrum in two independently sourced AutoDock Vina panels. Their participation-ratio (PR) effective dimensions were 1.83 and 2.27; removing ligand and target main effects changed the estimand and increased them to 9.30 and 18.21, while three target-wise independence nulls remained near their algebraic maxima. An observed-pair ChEMBL benchmark compared absolute, column-standardized and residual-scaled Vina scores without activity imputation. Their accuracies were 0.566, 0.546 and 0.564, and the residual-minus-absolute interval was -0.065 – 0.063 ; neither a ranking gain nor post hoc equivalence was resolved. A human binding K_i/K_d estimand reached the same precision-limited conclusion. Before target-specific scaling, removal of the ligand row effect was exactly rank-invariant, so operational differences arose from scale choices rather than centering alone. Two-way centering therefore diagnoses correlated residual variation; use of the scaled representation for target ranking still requires external task-level validation.

Scientific Contribution. We provide a reusable spectral-audit protocol that separates column-standardized and main-effect-free estimands, chemical and target composition uncertainty, and parametric and empirical null expectations. A ranking decomposition and endpoint-aligned ChEMBL benchmark show why higher residual dimension is a matrix diagnostic rather than evidence that a scaled residual score ranks targets better.

Keywords: molecular docking; AutoDock Vina; effective dimension; spectral analysis; two-way centering; reverse docking; target profiling.

1 Introduction

A docking score is scalar for one protein–ligand pose, but many applications produce a matrix. Reverse docking ranks targets for a ligand, secondary-pharmacology panels compare off-target liabilities, and multiobjective design uses scores against several desired and undesired proteins. In each case the useful object is a matrix $X \in \mathbb{R}^{N \times P}$ whose rows are ligands and columns are targets. Interpreting its P columns as P independent target-resolved descriptors assumes that the matrix actually contains a commensurate number of independent directions.

Two long-recognized biases make this assumption doubtful. Empirical energy functions often reward larger ligands because additional atoms contribute additional favorable contacts; normalization by molecular weight, heavy-atom count or learned physicochemical corrections can change virtual-screening rankings.[4, 5, 6] Across proteins, scores also depend on pocket contact area, size and hydrophobicity. In a reverse-docking study of 5,000 molecules and 100 proteins, these target properties produced recurrent “interference proteins” and motivated per-target normalization.[7] These observations identify ligand and target main effects. They do not say how much of a complete score matrix those effects occupy, how many collective directions remain, or whether a low-dimensional column-standardized surface is equivalent to poor selectivity.

Operational target-fishing studies reinforce the need to separate normalization from validation. A 13-procedure benchmark found strong target-dependent variation and evaluated consensus docking, while CRDS combined min–max-normalized GOLD, Vina and LeDock scores over 5,254 binding sites.[25, 26] Large external reverse-screening experiments also show that target retrieval depends on the amount and diversity of bioactivity knowledge, and recent open-source reverse-docking systems combine Vina with alternative or corrected scores rather than treating one raw energy scale as universal.[27, 28] These studies evaluate retrieval systems. Our analysis instead asks what the score matrix contains before a target-ranking rule is chosen and which transformations can change that rule.

Two-directional normalization and principal-component analysis of docking matrices also precede this study. SePreSA applied a ligand-wise Z transformation followed by normalization of protein-score vectors to a 79-compound by 86-protein docking matrix. [1] Fukunishi and

colleagues applied PCA to a complete docking matrix for compound and pocket classification, and later used multi-protein docking-score profiles as QSAR descriptors.[2, 3] These studies establish that matrix-wide normalization and PCA are useful. The contribution here is narrower: large-panel quantification of spectral concentration, explicit separation of column-standardized and two-way-centered estimands, chemical- and target-composition uncertainty, transformation-matched nulls, and a reusable audit protocol connecting these diagnostics to an operational ranking check.

The distinction matters because docking has several separable tasks. A method may reproduce a crystallographic pose yet rank affinities poorly, and no single program performs uniformly across targets and evaluation objectives.[8] Likewise, the aggregate variance of a score matrix and the correctness of within-ligand target contrasts are different objects. Writing

$$X_{ij} = \mu + a_i + b_j + \gamma_{ij}, \quad (1)$$

separates the grand mean μ , the ligand tendency a_i to score well across targets, the target tendency b_j to score ligands favorably, and the interaction γ_{ij} . The column-standardized spectrum is allowed to be dominated by a_i . Selectivity, however, depends on differences within rows and ultimately on γ_{ij} ; the additive terms cancel from a double difference. A PR estimate on X and the same estimate on γ therefore answer different questions. Established panel selectivity summaries such as Gini, entropy and window scores also answer different questions and can reorder the same compounds, making the metric definition part of the estimand rather than a cosmetic choice.[23, 24]

Complete public score matrices now permit this distinction to be tested at useful scale. DOCKSTRING contains docking scores for more than 260,000 molecules against 58 medically relevant targets.[9] An independent acute-toxicity resource contains 12,651 analyzable compounds docked against a 44-protein secondary-pharmacology panel.[10] We combine these matrices with matched experimental affinity data. Two small RF-Score rescoring blocks and learned-affinity examples are retained as illustrative sensitivities in the Supplement because neither provides a large independent replication of the principal Vina result.[11]

Low-rank representations are not new to chemogenomics. Kernelized Bayesian matrix factorization, probabilistic matrix factorization and neighborhood-regularized latent-factor models deliberately embed sparse DTI observations in lower-dimensional spaces to improve prediction or

matrix completion.[17, 18, 19] Our question is different: we do not impose a low-rank predictive model on sparse experimental data. We measure the empirical spectrum of dense docking output, then ask which part is attributable to additive ligand and target effects. The two-way decomposition connects this diagnostic to classical additive-model reasoning, while participation and entropy ranks provide continuous summaries rather than a chosen latent dimension.[20, 21]

We ask four linked questions. Do the two large tested Vina panels concentrate their variance into a few PR effective dimensions? Is that observation stable to chemical and target-panel composition? What remains after ligand and target main effects are removed? Finally, which parts of the residual transformation can alter within-ligand target order, and how precisely can their task-level effects be estimated on matched experimental data? The last comparison is necessary because a transformation can clarify an estimand without becoming a better scoring rule.

2 Results

2.1 Two large Vina panels have concentrated spectra

The 12,651-by-44 antitarget panel was analyzed after clipping the rare positive Vina scores to zero, mean-imputing missing entries and standardizing each target column. The largest eigenvalue accounted for 73.3% of the correlation spectrum. The PR effective dimension,

$$r_{\text{PR}} = \frac{(\sum_k \lambda_k)^2}{\sum_k \lambda_k^2}, \quad (2)$$

has a direct correlation interpretation. Because a P -target correlation matrix has $\text{tr}(C) = P$,

$$r_{\text{PR}} = \frac{P^2}{\text{tr}(C^2)} = \frac{P}{1 + (P-1)\overline{r_{ij}^2}}, \quad (3)$$

where the bar is over unique off-diagonal target pairs. Thus, a small PR dimension is an alternative summary of strong mean-squared inter-target correlation. PR adds no statistical information beyond mean r^2 for a correlation matrix; its practical role is to express the same dependence on a bounded scale of effective components. Mean $|r|$ gives a familiar correlation scale, while the full spectrum and correlation matrix retain signs, localization and the distribution of dependence among collective modes (Table 1). The estimated PR dimension was 1.834, with a Butina-cluster bootstrap interval of 1.783–1.884 (Fig. 1). Three leading modes exceeded both the descriptive

rank-wise 95% column-permutation threshold and a studentized maximum-deviation envelope controlling the family-wise error rate over eigenvalue ranks. The matrix is not algebraically rank two: eight principal components are needed to reach 90% of variance, and the entropy effective rank is 3.90. “Collapse toward two” therefore refers specifically to participation-ratio variance concentration. Full column-standardized and residual scree plots are given in Supplementary Fig. S1.

The 111 zero-censored cells did not drive this concentration. Treating them as missing and target-mean imputing gave a PR dimension of 1.829, while excluding every affected ligand gave 1.814 (12,544 ligands). Excluding the four most-censored targets gave 1.602, but this changes the target estimand and is interpreted as panel-composition sensitivity rather than an imputation effect (Supplementary Table S2). Complete-case analysis gave 1.320 on 9,297 ligands; because this selects a different chemical population, it is a support sensitivity, not a like-for-like missing-data correction. Although only 1.466% of cells were missing, 3,354 ligands had at least one missing value, usually for one target (median one missing cell; 90% range 1–10). Missingness was highly concentrated: target 4mq_s alone was absent for 3,352 ligands. Excluded ligands were also larger than complete cases (median molecular weight 399 versus 249 Da; median heavy-atom count 28 versus 17), with standardized mean differences of 1.86 and 2.03. The PR shift therefore accompanies substantial chemical- support selection rather than the nominal cell fraction alone (Supplementary Table S2 and the machine-readable support table). A chemical-cluster block-permutation null and a Marchenko–Pastur edge based on the 8,150 effective clusters both retained three leading modes, matching the independent-column null.

The independent DOCKSTRING matrix gave the same qualitative result at much larger N and with 58 targets. Among 260,060 molecules with complete scores, principal component 1 accounted for 65.9% of the spectrum and r_{PR} was 2.266 (molecule-bootstrap interval 2.239–2.298). Five independently seeded 15,000-molecule supports gave point estimates from 2.252 to 2.294; support-specific Bemis–Murcko scaffold-cluster intervals are shown in Supplementary Fig. S6. Four leading modes exceeded both the rank-wise and simultaneous common-protocol null thresholds. Its entropy rank was 6.21 and 22 components were needed for 90% of variance, again distinguishing a concentrated spectrum from literal rank deficiency. Retaining the 5,393 positive scores rather than clipping them changed the raw PR dimension only from 2.266 to 2.275. These intervals and support comparisons remain conditional on the fixed 58-target panel.

The result was also stable to the observed target-panel composition. Leave-one-target-

Table 1: Complementary summaries of target dependence. PR and mean squared off-diagonal correlation are algebraically equivalent for a correlation matrix; mean absolute correlation retains a more familiar correlation scale, while PC1 reports concentration in only the leading mode.

Matrix	surface	PR	mean r^2	mean $ r $	PC1
Docking-44	column-standardized	1.83	0.535	0.694	0.733
Docking-44	residual	9.30	0.087	0.246	0.274
DOCKSTRING-58	column-standardized	2.27	0.431	0.622	0.659
DOCKSTRING-58	residual	18.21	0.038	0.155	0.178

out PR dimensions ranged from 1.756 to 1.853 for Docking-44 and from 2.218 to 2.311 for DOCKSTRING-58 (Supplementary Fig. S2). Random target subsets showed saturation rather than a linear increase: at 20 targets the median column-standardized PR dimensions were 1.79 and 2.18, compared with 1.83 and 2.27 for the full panels (Fig. 1c). Broad target-family-stratified sampling gave similar 20-target medians of 1.85 and 2.22. For the two-way-centered residuals, the corresponding family-stratified medians were 7.38 and 11.40, increasing to 9.30 and 18.21 in the full panels (Fig. 1d). These subset distributions quantify sensitivity within the tested biological panels; the target lists are not probability samples from a universal target population.

2.2 The leading direction is a shared docking-score axis

The dominant direction was shared across targets rather than localized to one receptor family. After orienting principal component 1 (PC1) consistently, 43 of 44 Docking-44 loadings and 57 of 58 DOCKSTRING loadings were positive (Fig. 2c,d). The PC1 scores correlated 0.997 and 0.995 with the corresponding per-ligand mean of the original score matrix. These are raw score-scale row means, not means calculated after column standardization. Seven physicochemical descriptors explained 60.6% of the variance in the Docking-44 row mean; heavy-atom count alone explained 18%.

Removing size-related information weakened but did not erase the concentration. Linear residualization of each target score against seven physicochemical descriptors increased r_{PR} from 1.83 to 2.54; cross-fitted nonlinear residualization increased it to 3.52. Restricting the data to 4,308 molecules with 17–22 heavy atoms still gave PR dimension 1.70. Dividing scores by heavy-atom count made the spectrum more concentrated (PR dimension 1.14), showing that ligand-efficiency normalization is not a dimensionality correction. Across 36 targets, pocket volume correlated 0.514 (target bootstrap interval 0.272–0.686) with the absolute correlation between each target and the per-ligand panel mean. A 13-descriptor pocket model had negative

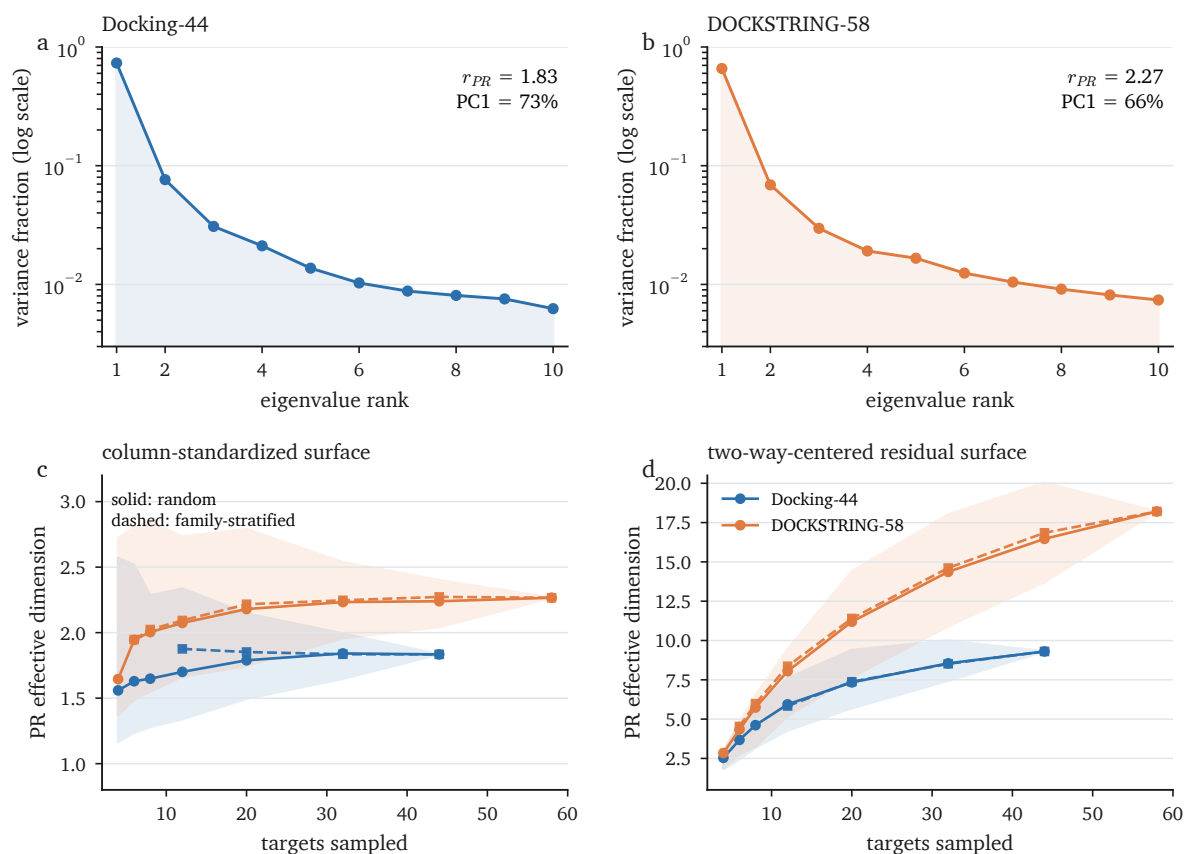


Figure 1: Two independent Vina panels have concentrated column-standardized spectra. a,b, Eigenvalue spectra and PR effective dimensions of the 44-target antitarget and 58-target DOCKSTRING matrices after column standardization. **c,d,** Column-standardized and two-way-centered residual PR effective dimensions under target subsampling. Solid lines and shaded 2.5–97.5% intervals show 200 uniform subsets; dashed square-marked lines show 200 family-stratified subsets where the requested size permits representation of every broad family. Full-panel points use all ligands.

leave-one-target-out R^2 for both ordinary least squares (-0.011) and ridge regression (-0.144), so the data do not support a multivariable pocket-mechanism claim. On the exact 74-by-6 experimental support, linear and five-fold cross-fitted nonlinear descriptor residualization raised docking PR dimension only to 2.81 and 2.68, respectively, still below the exact-relation median experimental value of 3.83. Thus the matched contrast is not solely a consequence of the seven measured bulk descriptors.

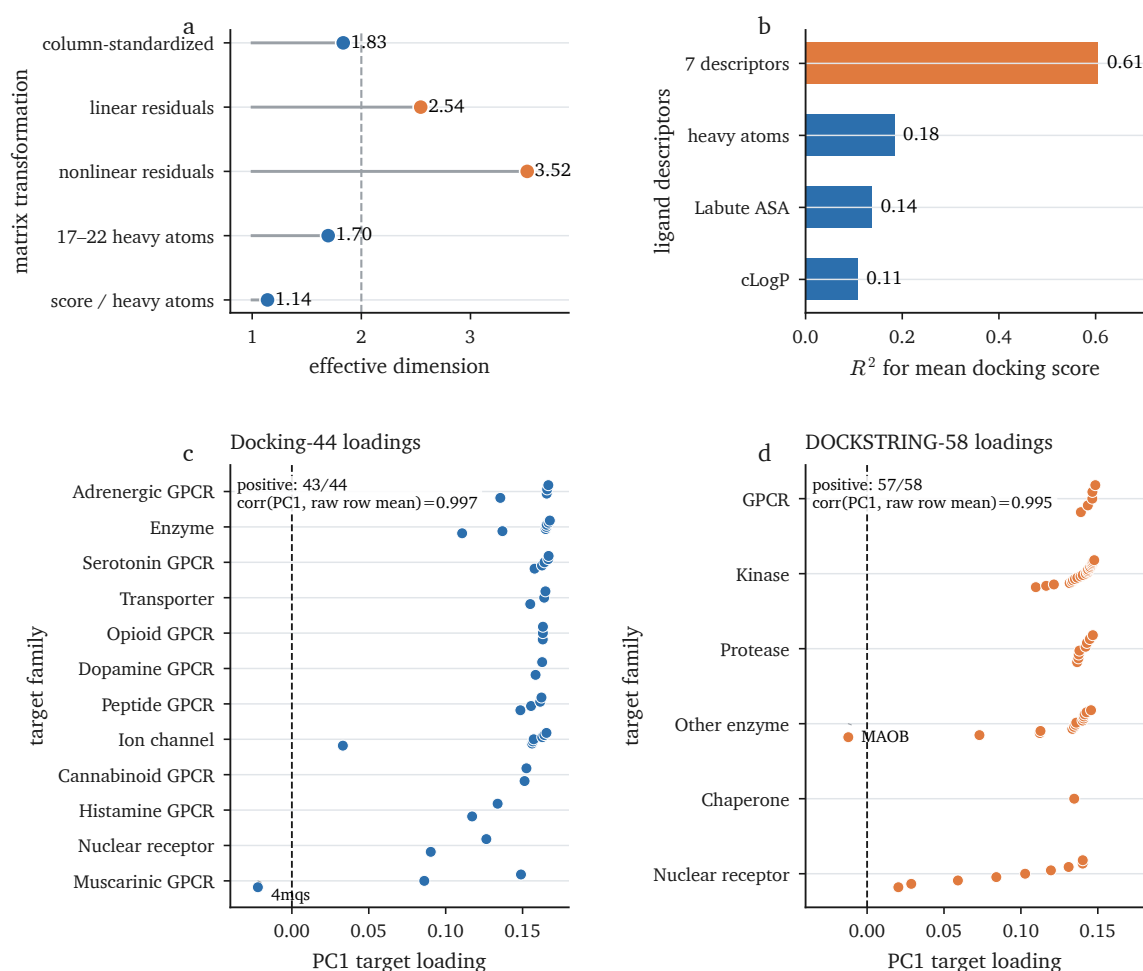


Figure 2: The leading component is a shared docking-score axis. **a**, Docking-44 PR effective dimension after physicochemical transformations; the horizontal segments connect each estimate to one and are visual guides, not confidence intervals. **b**, Variance in the raw per-ligand mean score explained by seven ligand descriptors and single-descriptor controls. **c,d**, PC1 target loadings grouped by curated target family. Loading signs were oriented so that the majority are positive. The annotations report the number of positive loadings and the correlation between PC1 scores and raw per-ligand mean scores. Descriptor residualization is diagnostic and is not proposed as a scoring function.

2.3 The docking–experiment contrast narrows after centering

A concentrated docking surface could reflect its compounds or targets rather than its score model. We therefore used the same 74 compounds and six aminergic targets for docking and ChEMBL. The primary experimental curation required an exact standard relation, retained Ki, Kd, IC50 and EC50, aggregated repeated compound–target activities by the median, and required measurements for at least five of six targets. Target-median imputation filled 9.9% of cells. The resulting ChEMBL surface had PR dimension 3.833, compared with 1.827 for docking on the same support (Fig. 3a). The paired ligand-bootstrap interval for the experimental-minus-docking difference of 2.006 was 1.174–2.673.

The comparison changed in the residual space. After two-way centering, the docking and experimental PR dimensions were 3.646 and 4.128. Their difference was 0.483, with a paired bootstrap interval of -0.071 – 0.921 (Fig. 3b). Thus, much of the raw docking–experiment contrast on this support arose from the shared ligand direction. It did not persist as a clearly resolved difference between residual effective dimensions.

Multiple imputation supported the same qualitative reading. Across 100 posterior chained-regression imputations, the median experimental PR dimensions were 3.823 before centering and 4.069 afterward. The corresponding experimental-minus-docking differences had medians 1.996 and 0.423. The displayed 95% ranges across imputations quantify missing-value sensitivity; they are not bootstrap confidence intervals.

More restrictive assay blocks preserved the ordering. Human binding assays with all four endpoints gave 3.510 for ChEMBL and 1.865 for docking on 47 ligands. Restricting further to human binding Ki/Kd gave 3.072 and 1.894 on 35 ligands. These comparisons do not define the dimension of experimental binding in general: the assays come from multiple laboratories and conditions, up to 13.3% of cells are imputed, and six targets cap the centered algebraic dimension at five. They support only the statement that these matched ChEMBL surfaces are less spectrally concentrated than docking on the same ligand–target support.

Additional controls addressed assay choice, missingness and measurement heterogeneity. A fully observed 31-by-6 ChEMBL sub-block had PR dimension 3.61. On a secondary 212-by-22 block, estimates were 5.72 with K-nearest-neighbour imputation, 8.59 averaged over chained imputations, 2.64 with low-rank matrix completion and 7.28 with probabilistic-PCA expectation maximization; the magnitude depended on the estimator. A separate frozen ChEMBL extraction with the same nominal 212-by-22 shape and 52.9% missingness gave KNN PR dimension 6.85; we

retain both in Supplementary Table S3 as a curation-sensitivity discrepancy. A legacy most-potent aggregation of the 74-by-6 block gave 3.734 for experiment and 1.827 for docking, with a paired bootstrap difference interval of 1.05–2.61, and is retained only as a sensitivity. Different-document repeats estimated a standard deviation of 0.493 pChEMBL units. Adding target-scaled Gaussian noise at that level to docking raised its median PR dimension to 2.407 (simulation interval 2.21–2.64); approximately 2.5 times that scale was required to reach the legacy experimental value. None of these controls removes all biological, laboratory or condition-related heterogeneity.

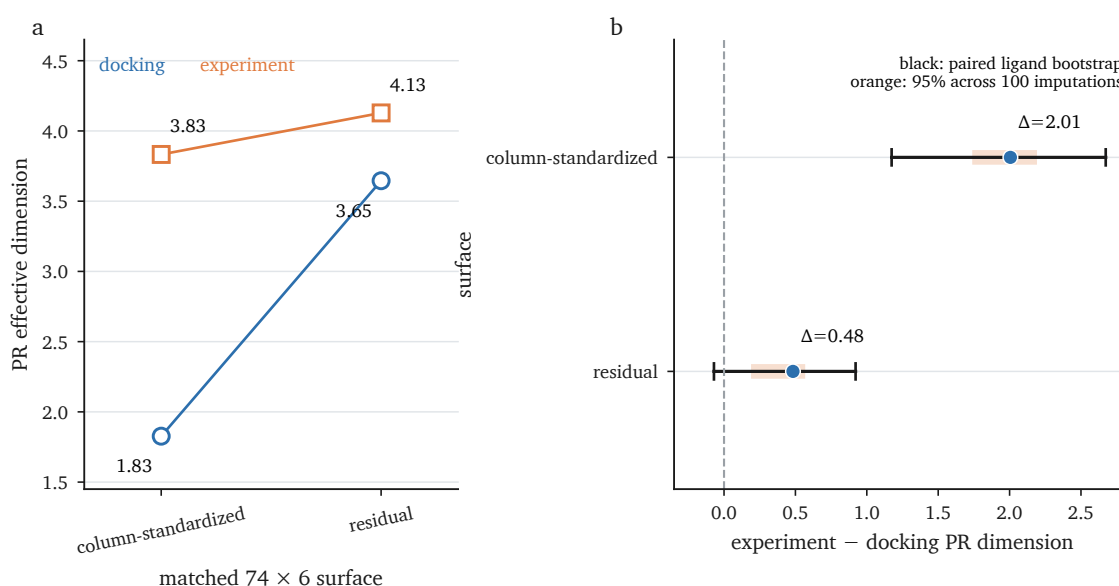


Figure 3: The matched docking-experiment contrast is mainly a raw-surface contrast. **a**, PR effective dimensions for docking and exact-relation median ChEMBL on the same 74 compounds and six targets, before and after two-way centering. **b**, Experimental-minus-docking differences. Black intervals are paired ligand-bootstrap 95% intervals; orange bands are the central 95% of the difference across 100 posterior multiple imputations. These intervals answer different uncertainty questions and are not pooled.

2.4 Centering reveals higher-dimensional but still concentrated residual surfaces

The column-standardized spectrum mixes the ligand main effect with the ligand-target residual. We removed both row and column means according to Eq. 1 and then applied the same column standardization and spectral estimator. This changed the measured object and the answer (Fig. 4a). Docking-44 rose from PR dimension 1.834 to 9.303; a paired Butina-cluster bootstrap placed the increase at 6.982–7.909. DOCKSTRING rose from 2.266 to 18.206, and five independent 15,000-molecule supports gave increases from 15.880 to 16.045. The corresponding PC1 fractions fell from 73.3% to 27.4% and from 65.9% to 17.8%. In unstandardized sum-of-squares terms, the residual retained 27% and 31% of grand-centered variation. Low variance relative to

a main effect is not absence.

The residual dependence was distributed across positive and negative target pairs rather than forming another uniformly signed axis. Mean absolute off-diagonal target correlation fell from 0.694 to 0.246 in Docking-44 and from 0.622 to 0.155 in DOCKSTRING; the central 90% residual-correlation ranges were -0.450 – 0.500 and -0.328 – 0.333 (Supplementary Fig. S7). Within-family absolute correlations exceeded between-family values by only 0.030 and 0.017, and broad-family label permutations did not resolve these small differences ($p = 0.073$ and 0.067). The leading residual modes instead remained strongly associated with ligand chemistry: residual-PC1 scores had absolute Spearman correlations $|\rho| = 0.72$ and 0.77 with molecular weight or heavy-atom count; component signs are arbitrary. Removing the common ligand offset therefore leaves target-specific physicochemical slopes; the evidence does not identify the residual modes as protein-family or biophysical specificity axes.

The increase was not only a rescaling artifact of the PR denominator. Relative to the available algebraic maximum, normalized PR dimensions increased from 0.042 ($1.834/44$) to 0.216 ($9.303/43$) for Docking-44 and from 0.039 ($2.266/58$) to 0.319 ($18.206/57$) for DOCKSTRING. This normalization is a ceiling reference, not a correction for target count: the residual PR dimension increased with panel size while its ratio to $P - 1$ decreased (Fig. 1d). At a matched 44 targets, the DOCKSTRING median residual PR dimension was 16.47 (target-subset interval 13.72–20.03), compared with 9.30 for the full Docking-44 panel. The cross-dataset difference therefore remained at matched P , although part of the 18.21-versus-9.30 contrast reflects different target counts.

Higher-dimensional did not mean unconcentrated. A fitted additive Gaussian null retained the observed grand mean, ligand effects, target effects and target-specific residual variances, but drew residuals independently. Its median residual PR dimensions were 42.40 for Docking-44 and 56.48 for DOCKSTRING, close to the respective $P - 1$ ceilings of 43 and 57. A nonparametric companion independently permuted each observed residual target column, sampling from its empirical values without a Gaussian assumption; its medians were also 42.40 and 56.48. A third null projected independent random directions into the zero-row-sum subspace and rescaled every simulated row to its observed residual norm. It preserved the ligand-specific residual magnitude exactly and gave medians 42.63 and 56.62. The observed values, 9.29 and 18.16 on the same 12,000-row supports, were smaller than every one of 500 realizations under either null (one-sided empirical $p = 0.002$; Fig. 4b). Chemical redundancy did not account for this

separation: across 200 supports containing one ligand from each of 8,150 Docking-44 Butina clusters, the median observed residual PR was 9.43 and the matched empirical-null median was 42.33. The empirical null removes the Gaussian assumption, while the random-direction null addresses ligand-specific residual heteroskedasticity. Together they show residual dependence far stronger than target-wise independent expectations.

In the transformation-matched permutation analysis, the descriptive rank-wise mode counts increased from 3 to 7 and from 4 to 10. The same counts survived the simultaneous studentized maximum-deviation envelope and were unchanged in five independent 100- permutation series. Because even weak correlations can become statistically detectable at large sample size, we use these counts to describe spectrum shape rather than as a measure of practical dimension.

Two-way centering also creates an exact structural constraint: every interaction row sums to zero. Non-zero column scaling preserves one linear dependence, so the interaction matrix has algebraic rank at most $P - 1$ and one correlation eigenvalue is zero. Residual and column-standardized PR dimensions therefore describe different constrained spaces; the centered increases occur despite, not because of, a larger algebraic maximum.

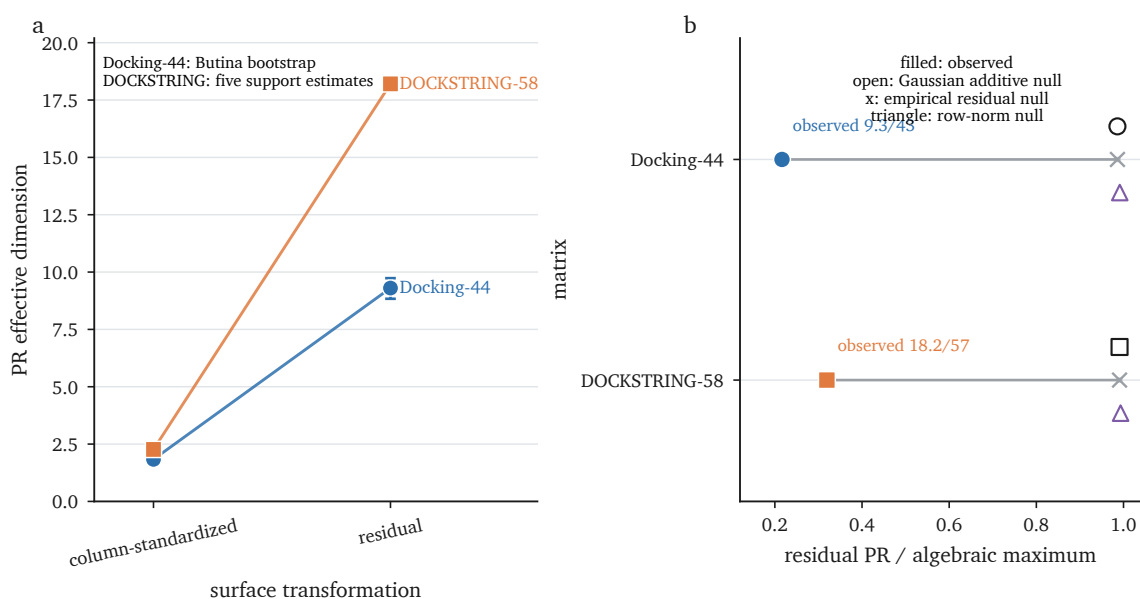


Figure 4: Two-way centering increases effective dimension without removing residual concentration. **a**, Paired raw-to-residual changes. Docking-44 error bars are 95% Butina-cluster bootstrap intervals; DOCKSTRING error bars span point estimates across five independently seeded 15,000-molecule supports. **b**, Observed residual PR dimensions and null medians, each divided by its $P - 1$ algebraic maximum. Open markers and upper bars show the fitted Gaussian additive null; crosses show the nonparametric residual-permutation null; open triangles and lower bars show the row-norm-preserving random-direction null. Each bar is the central 95% of 500 realizations and is visually narrower than its marker. All nulls remove cross-target residual correspondence; the last additionally preserves every observed ligand-specific residual norm before column standardization.

2.5 Only target-specific scaling changes target order

We expanded the operational endpoint beyond the dense six-target block. Two complementary estimands were declared together during revision rather than prospectively registered: an all-endpoint analysis maximizing coverage and a human binding Ki/Kd analysis reducing assay heterogeneity. Exact full-InChIKey matching identified 276 Docking-44 ligands with at least one exact-relation Ki, Kd, IC50 or EC50 measurement in ChEMBL. Requiring at least two observed targets retained 137 ligands, 38 targets, 691 experimental cells and 2,522 non-tied within-ligand target pairs. No experimental activity was imputed. The primary mean gave every ligand equal weight; predicted score ties received half credit, and its 95% intervals resampled either 95 Bemis–Murcko scaffold clusters or 121 fingerprint-based Butina clusters. All docking offsets, scales and imputation means were fitted after excluding the 137 evaluation ligands from the 12,651-ligand reference panel; 89 of their 6,028 docking scores were imputed from these external target means.

The ranking effect admits a useful algebraic decomposition. Let m_j be the external mean for target j , r_i the mean of evaluation ligand i over all 44 targets and g the external grand mean. Before target-specific scaling, target-centered scores $C_{ij} = X_{ij} - m_j$ and two-way-centered scores $R_{ij} = X_{ij} - m_j - r_i + g$ obey

$$R_{ij} - R_{ik} = C_{ij} - C_{ik}. \quad (4)$$

Thus row centering alone cannot change any within-ligand target order. The two unscaled representations consequently had exactly the same accuracy, 0.556. Column standardization, target centering followed by residual target-scale standardization, and full two-way-residual standardization reached 0.546, 0.549 and 0.564, respectively. Division by target-specific scales makes the term $(-r_i + g)/s_j$ target dependent; any ranking change attributed to the operational residual representation therefore arises from the interaction between this ligand offset and target-specific scaling, not from unscaled row centering itself.

Absolute Vina reached mean per-ligand pairwise target-preference accuracy 0.566 (Murcko-cluster interval 0.502–0.623; Fig. 5a). The paired residual-minus-column difference was 0.017 (union of cluster-bootstrap 95% intervals, -0.022 – 0.055), and the residual-minus-absolute difference was -0.002 (-0.065 – 0.063). Conservative unions of the corresponding 90% intervals were -0.015 – 0.049 and -0.056 – 0.051 . In post hoc equivalence sensitivities neither contrast met a ± 0.02 margin; residual minus absolute also failed a ± 0.05 margin, while residual minus

column met only that more permissive margin. These margins were not prospectively registered. The benchmark therefore excludes neither moderate benefit nor moderate harm and does not establish equivalence among the representations; its estimated minimum detectable difference was approximately 0.054. Ordinary ligand-identity permutations placed the absolute and residual results above their nulls ($p = 0.014$ and 0.024), but that evidence did not survive chemical redundancy: across 25 one-ligand-per-Murcko-cluster supports, median permutation probabilities were 0.134 and 0.284. Within-ligand permutation of the experimental order gave $p = 0.009$ and 0.012 . These tests assess score–experiment association, not superiority of one representation, and the cluster-aware sensitivity weakens the inclusive association claim.

Target deletion exposed an additional uncertainty axis. The residual-minus-absolute difference ranged from -0.030 to 0.010 across the 38 broad-panel leave-one-target-out analyses; for residual minus column-standardized it ranged from 0.002 to 0.029 . These are composition sensitivities conditional on the observed targets. The target panel was not randomly sampled, so neither the ligand/scaffold bootstrap nor the target jackknife licenses superpopulation inference to unmeasured targets.

Target priors exposed a more consequential boundary. A target order fitted on the 12,514 non-evaluation docking ligands reached 0.525, and an external ChEMBL target prior fitted after excluding the evaluation ligands reached 0.525. Their point estimates were below residual Vina, although the paired residual-minus-prior intervals included zero. By contrast, a target prior within the broadly profiled evaluation cohort, fitted with the held-out ligand’s entire scaffold cluster excluded, reached 0.592 (0.531 – 0.654). It exceeded residual Vina by 0.028 on average, but the paired interval (-0.127 – 0.066) was wide. This cohort prior is a supervised observation-process diagnostic, not a deployable target-fishing score.

Coverage sensitivity changed the score-based ordering (Fig. 5b). Requiring at least three observed targets retained 93 ligands: absolute Vina reached 0.587 (0.537 – 0.638), residual Vina 0.530 (0.480 – 0.574), the docking target prior 0.590 (0.542 – 0.637) and the cohort prior 0.636 (0.576 – 0.693). At thresholds of five and six observed targets, absolute accuracy remained 0.581 and 0.587 while residual accuracy was 0.540 and 0.537. These nested, smaller strata have broad pointwise intervals, but they show that the inclusive ordering depends in part on the 44 ligands observed for exactly two targets.

The assay-restricted estimand reached the same precision-limited conclusion. On 107 ligands and 30 targets with human binding K_i/K_d measurements, absolute, column-standardized and

residual accuracies were 0.565, 0.554 and 0.579 over 2,224 pairs; residual minus absolute had a cluster-interval union of $-0.051-0.085$. Its leave-one-target-out difference ranged from -0.027 to 0.031 . The positive residual point shift was therefore unresolved and target-composition sensitive.

Holding the broad 137-by-38 support fixed but comparing only cells with the same endpoint type produced 4,330 endpoint-specific instances across 132 evaluable ligands. Residual accuracy was 0.563 for Ki, 0.524 for Kd, 0.516 for IC50 and 0.398 for EC50; the Kd and EC50 estimates used only seven and 18 evaluable ligands. Because a target pair could occur under more than one endpoint, a duplicate-collapsed sensitivity averaged each ligand-target-pair once after endpoint-specific scoring. It retained 2,460 unique instances and gave absolute, column-standardized and residual accuracies 0.567, 0.541 and 0.562, again without a resolved residual-minus-absolute difference. Removing cells whose repeated measurements spanned more than one pChEMBL unit gave residual accuracy 0.560 on 1,912 pairs (Supplementary Table S5 and the accompanying machine-readable tables).

This comparison supplies the operational boundary of the spectral analysis. Two-way centering reveals whether ligand and target main effects conceal correlated residual variation, but unscaled row centering cannot change target order and residual dimension does not determine ranking accuracy. A reverse-docking workflow should decompose centering from scaling, report target-prior and ligand/outcome-permutation baselines, stratify by assay type and experimental coverage, and choose its score representation by external task-level validation rather than by effective dimension alone.

3 Discussion

The two large Vina panels share a concentrated column-standardized spectrum, but centering does not make the residuals unconcentrated. The PR dimensions of 1.83 and 2.27 rose to 9.30 and 18.21 after ligand and target main effects were removed; fitted Gaussian additive nulls and both empirical-marginal and row-norm-preserving nulls placed target-wise independent residuals near 43 and 57 dimensions. The empirical residuals therefore contain more directions than the shared-score surface while remaining strongly correlated. The result is a reduction in spectral concentration, not its disappearance.

Two-way normalization and PCA themselves predate this work. SePreSA used two-directional

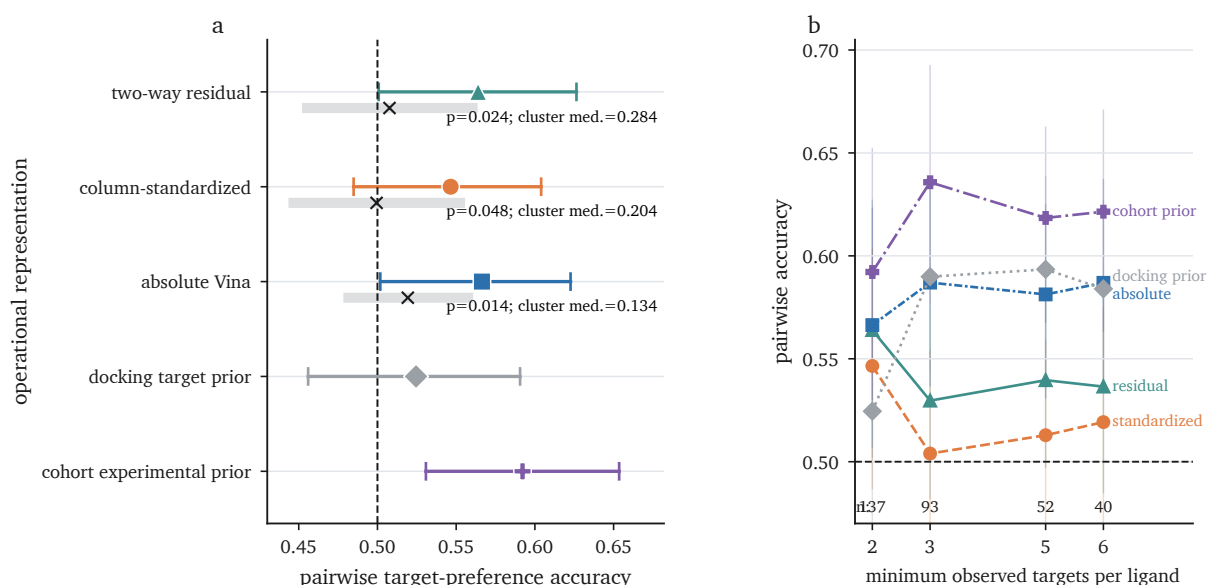


Figure 5: Spectral and operational conclusions are distinct. **a**, Mean per-ligand pairwise target-preference accuracy over 2,522 observed, non-tied experimental target pairs from 137 ligands and 38 targets. Colored intervals are 95% Bemis–Murcko scaffold-cluster bootstrap intervals; the independent Butina-cluster intervals are tabulated in Supplementary Table S5. Grey bars and crosses show the central 95% and mean of 5,000 ligand-identity permutations for score-based representations; labels give the ordinary one-sided probability and the median probability across 25 one-ligand-per-Murcko-cluster supports. The cohort experimental prior is fitted with the evaluation ligand’s entire scaffold cluster held out and is a supervised observation-process diagnostic rather than a deployable docking score. **b**, Accuracy after requiring each ligand to have measurements for at least two, three, five or six targets; vertical lines are pointwise 95% scaffold-bootstrap intervals and labels give the number of retained ligands. The strata are nested rather than independent. The dashed line is chance.

normalization of docking-score matrices, and earlier protein–compound affinity work used principal components as multidimensional descriptors.[1, 2, 3] Our contribution is the quantitative separation of surface and residual spectral estimands on two large dense panels, uncertainty with respect to target and chemical composition, multiple structure-matched nulls, and an operational decomposition of how centering and scaling affect target ranking. We consolidate these elements as a reusable spectral-audit protocol rather than present centering, PCA or PR as a new algorithm. PR is exactly a monotone re-expression of mean squared target correlation; its advantage is interpretive scale, while the correlation matrix and full spectrum remain necessary for signs, localization and mode shape.

The loading analysis explains why the estimands diverge. Almost every target loaded in the same direction on PC1, and PC1 scores were nearly identical to the raw per-ligand mean score. Ligand bulk accounted for part, but not all, of this direction. Column standardization removes target offsets while retaining the shared ligand tendency; two-way centering removes that tendency and reveals the remaining cross-target covariance. We call this a shared docking-score axis rather than a general-binding axis because neither its biological meaning nor universal validity was established.

The residual covariance also has a concrete, if incomplete, interpretation. Its leading modes remain associated with molecular size, consistent with target-specific size slopes that survive removal of a common ligand offset. Broad protein-family labels explain at most a weak part of the residual correlation pattern on these panels. We therefore use “correlated residual variation” rather than equating it with biological specificity; sequence- or pocket-level attribution remains open.

On the matched ChEMBL support, experimental affinities were less concentrated than docking before centering, but the paired residual-space difference was smaller and its bootstrap interval included zero. This does not show that Vina residuals reproduce experimental geometry. It shows that the raw docking–experiment contrast on this support is largely a contrast in shared main effects rather than a resolved difference in residual PR dimension.

Operational validation changed the interpretation again. Unscaled target centering and unscaled two-way centering produce exactly the same within-ligand target order; only target-specific scaling can break this invariance. On the expanded 137-ligand, 38-target observed-pair benchmark, absolute and standardized two-way-residual Vina had nearly identical point estimates, and no paired score-transformation contrast was resolved. The uncertainty did not establish

equivalence even at a post hoc ± 0.05 margin for residual versus absolute scores. Ordinary identity and outcome permutations placed both above their nulls on the inclusive support, but one-ligand-per-scaffold permutations weakened the identity evidence, and the ordering shifted toward absolute Vina when ligands with only two observed targets were excluded. Fixed-support same-endpoint comparisons and the human binding K_i/K_d subset did not resolve a different conclusion. A scaffold-held-out experimental cohort prior was comparable or stronger but is a supervised diagnostic, not a deployable docking rule. Precisely because it reflects assay selection, it demonstrates how a ranking result can depend on which targets were measured. Higher residual dimension is therefore neither necessary nor sufficient for a useful reverse-docking score.

Accordingly, we propose a nine-step spectral audit: (1) define the score matrix, sign convention, missingness rule and chemical/target support; (2) compute both the column-standardized surface and the two-way-centered residual estimands; (3) report mean r^2 , mean $|r|$, PR and entropy effective dimensions, PC1 fraction, correlation distributions, cumulative-component counts and the algebraic maximum; (4) quantify chemical-support uncertainty with a scaffold or chemical-cluster bootstrap; (5) examine target composition with leave-one-target-out and family-stratified subsampling; (6) compare with transformation-matched parallel-analysis, additive Gaussian, empirical residual-permutation and row-norm-preserving nulls; (7) for ranking applications, test invariances and separate centering from target-specific scaling; (8) evaluate absolute, transformed and target-prior baselines with identity/outcome permutations and assay/coverage sensitivities; and (9) report paired uncertainty, any equivalence margin, and the exact scope of generalization. Supplementary Table S9 maps each step to the released implementation, and `analysis/spectral_audit.py` provides a generic command-line entry point for dense score matrices. Two-way centering remains useful diagnostically even when it is not selected as a ranking rule.

Both large matrices use Vina-family scores, and the small RF-Score blocks in Supplementary Fig. S3 cannot represent modern learned scoring functions. Neither target panel is a probability sample from all pharmacologically relevant proteins, so jackknife and family-stratified subsampling quantify stability within the observed panels. The dense docking-experiment spectral control still contains only 74 ligands and six aminergic targets, with 9.9% imputed experimental cells. The expanded operational benchmark covers 38 targets but is sparse and non-random: pairwise target co-observation has a median support of only three ligands, and its inclusive result includes 44 ligands observed against exactly two targets. Fixed-support same-endpoint and hu-

man binding K_i/K_d analyses reduce endpoint heterogeneity but also reduce information; neither produces a precise score-transformation contrast. Target deletion further shows that the paired differences depend on the fixed target composition. Scaffold resampling, measurement-dispersion filtering and coverage thresholds expose but cannot remove ChEMBL’s assay-selection and laboratory heterogeneity. The benchmark excludes a large advantage but is not precise enough to establish equivalence; it supports an observed-target ordinal claim, not unmeasured-target retrieval. Finally, the public package reproduces analysis from frozen matrices; it does not rerun upstream receptor preparation, docking, pose generation or RF-Score rescoring.

4 Conclusions

In the two tested large Vina matrices, two-way centering reveals substantially higher-dimensional correlated residual variation, but the residuals remain far more concentrated than Gaussian, empirical-marginal or row-norm-preserving target-wise independent nulls. On dense matched ChEMBL data, the transformation narrowed the docking–experiment PR contrast. On a broader observed-pair benchmark, standardized residual scores showed neither a resolved target-ranking advantage nor equivalence to absolute Vina; unscaled row centering was exactly rank invariant and the observed changes arose from target-specific scaling. The practical consequence is to separate diagnosis from correction: use the released spectral audit to identify shared and residual structure, then require any operational score transformation to outperform absolute, target-prior, permutation and assay/coverage-stratified baselines on held-out experimental data.

5 Methods

5.1 Datasets and score panels

The antitarget matrix was taken from the open resource of Nikitin et al.[10] and contained 12,651 analyzable compounds and 44 Vina-GPU target scores.[16] The source pipeline clipped positive scores to zero; its processed matrix contained 111 such zero-censored cells (0.020%) and no remaining positive values. Missing entries comprised 1.466% of cells and were replaced by the target mean for the primary analysis. Chemical-cluster uncertainty used the precomputed 8,150 Butina clusters supplied with the dataset. A three-target Docking-47 expansion is reported only as an illustrative sensitivity in the Supplement.

DOCKSTRING was read from its released full score table.[9] We used the 58 target columns, clipped positive scores to zero for consistency and used the 260,060 rows with complete scores for the primary spectrum. Only 0.0022% of entries were missing and 0.0358% were positive before clipping; an unclipped sensitivity retained these values.

RF-Score v1 was evaluated with the Open Drug Discovery Toolkit implementation.[11, 14] The first panel comprised the 41 ligands complete across RF-Score, smina/Vina and original Vina scores for 14 targets. The second comprised 35 ligands complete across RF-Score on Vinardo poses and original Vina scores for 20 targets. These are scoring-function and pose-source sensitivities, not affinity benchmarks, and are reported with the Docking-47 expansion in Supplementary Fig. S3.

Experimental activities came from ChEMBL release 34.[22] We retained records with a pChEMBL value and standard type Ki, Kd, IC50 or EC50; canonical SMILES were converted to full InChIKeys and targets were mapped to the panel's single-protein accessions. The primary matched block was rebuilt from activity-level records, required an exact standard relation and full InChIKey, and aggregated repeated compound–target activities by the median. It used 74 compounds with docking scores for six targets and ChEMBL measurements for at least five targets; 9.9% of experimental cells were median-imputed within target. We additionally restricted to human binding assays, either with all four endpoints or with Ki/Kd only. A secondary 212-by-22 block required at least eight measured targets per ligand and 30 measurements per retained target. We compared five-nearest-neighbor imputation, multiple imputation by chained equations, low-rank matrix completion and probabilistic-PCA expectation maximization. Assay sensitivity separately retained binding records with Ki or Kd endpoints from the already human-only curated refetch. Relations reported as inequalities and records without pChEMBL were excluded. A frozen table that retained the most potent pChEMBL value per compound–target pair was analyzed only as a legacy sensitivity with a paired ligand bootstrap.

For the expanded operational benchmark, the same exact-relation activity-level curation was matched by full InChIKey to all 12,651 Docking-44 ligands before imposing a dense-block criterion. Among 276 ligands with any matched activity, 137 had observations for at least two targets. These ligands contributed 691 observed cells across 38 targets; missing cells were left missing because the endpoint used only experimentally observed within-ligand target pairs. Restriction to human binding assays and then to human binding Ki/Kd defined assay sensitivities. The original 74-by-6 block was retained for the dense same-support spectral comparison and its

multiple-imputation sensitivity, not as the primary operational support.

DAVIS experimental affinity[13] and Boltz-2[12] were read from versioned matrices in the release only for a contextual centering ladder in Supplementary Fig. S4. The common surface contained 72 ligands and 442 kinases. PR dimensions were computed on each complete dense matrix, including pKd-floor values, because the surfaces were aligned. The exploratory rank-selectivity analysis is reported entirely in the Supplement; its complete panel has 20 arms, while the 12-arm above-chance subset is an outcome-restricted sensitivity.

5.2 Spectral quantities

For each complete matrix X , columns with zero variance were removed and the remaining columns were standardized to zero mean and unit sample variance, producing Z . We formed the target correlation matrix

$$C = \frac{1}{N-1} Z^T Z \quad (5)$$

and sorted its non-negative eigenvalues $\lambda_1 \geq \dots \geq \lambda_P$. The primary summary was the PR effective dimension in Eq. 2. We also computed the entropy rank $\exp[-\sum_k p_k \log p_k]$, where $p_k = \lambda_k / \sum_l \lambda_l$, the fraction explained by principal component 1 and the number of components reaching 80% and 90% cumulative variance. Equation 3 follows from $\text{tr}(C^2) = P + P(P-1)\overline{r_{ij}^2}$. The PR effective dimension is especially sensitive to dominant eigenvalues, whereas entropy rank weights the spectral tail more strongly. We additionally reported mean and median absolute off-diagonal correlation, the signed 5th–95th percentile range and the fraction of positive correlations. PR and mean squared correlation are algebraically equivalent; the other summaries and the full matrix retain information discarded by that scalar identity. Effective ranks are continuous summaries of variance concentration, not algebraic or nonlinear manifold ranks.

The Marchenko–Pastur upper edge $(1 + \sqrt{P/N})^2$ was used as an asymptotic guide. The original panel analyses anchored mode counts on column-wise permutation parallel analysis, which preserves each score marginal while breaking cross-target alignment.[15] For direct column-standardized-residual comparison, we additionally drew the same 12,000 rows from each large matrix, generated 500 independent column-wise permutations, applied either no centering or two-way centering to both observed and permuted matrices, and counted observed eigenvalues above the rank-wise 95th percentile of the corresponding null. Because rank-wise comparisons do not control multiplicity, we also formed a 95% simultaneous envelope from

the maximum studentized deviation over all eigenvalue ranks. Rank-wise counts are treated as descriptive; the simultaneous envelope controls the family-wise error rate. We repeated the count in five independent 100-permutation series to assess Monte Carlo stability. The 44-target analysis additionally used cluster-preserving and cluster-representative nulls; the latter retained three leading modes and controlled the inflated false-positive rate caused by chemical clustering. Effective-dimension intervals and the paired raw-to-residual difference for that panel resampled Butina clusters 500 times. DOCKSTRING used 100 molecule and 100 Bemis–Murcko scaffold-cluster replicates on each of five independently seeded 15,000-row supports. The primary matched experiment used a paired ligand bootstrap. These intervals are conditional on the observed target panels; chemical resampling does not represent target-selection uncertainty.

5.3 Main-effect removal

For complete matrices we computed the interaction residual in one closed-form step,

$$\Gamma = X - \bar{X}_{\cdot j} - \bar{X}_{i \cdot} + \bar{X}_{\dots}, \quad (6)$$

where the bars denote column, row and grand means. We then column-standardized Γ and computed exactly the same spectral summaries. Because standardization already removes column offsets, the change from the column-standardized to residual spectra is driven principally by removal of the per-ligand common direction. The fraction of grand-centered unstandardized sums of squares retained by Γ was reported separately and was not used as a PR-dimension measure. Each row of Γ sums to zero. Subsequent non-zero column scaling preserves one exact linear dependence, so $\text{rank}(\Gamma) \leq P - 1$ and the residual correlation spectrum contains one structural zero eigenvalue. For an additive model with independent, identically distributed residual noise, the non-zero residual spectrum is flat and the PR dimension approaches $P - 1$. We therefore treat $P - 1$ as a noise ceiling, not as a panel-size correction, and show target-subsampling curves for the residual estimand.

To test whether centering alone mechanically produced the observed residual spectra, we fit the additive model in Eq. 1 on a common 12,000-row support. For each of 500 realizations, residuals were drawn independently from zero-mean Gaussian distributions with the observed target-specific residual variances, then added to the fitted grand, ligand and target effects. We two-way centered each simulated matrix and recomputed PR dimension. The one-sided empirical

probability was calculated with the finite-simulation correction $(1 + k)/(1 + 500)$, where k is the number of null PR dimensions no larger than the observed value.

We also used a nonparametric residual-permutation null. Starting from the observed two-way-centered residual matrix on the same support, each target column was independently permuted across ligands, after which two-way centering and column standardization were repeated. The permutation step therefore sampled exactly from every target’s empirical residual values, including skewness and tails, while removing cross-target residual correspondence; transformation-matched re-centering can modify the resulting marginals slightly. This null does not preserve ligand-specific residual scale. For Docking-44, 200 additional realizations selected one ligand from each of the 8,150 Butina clusters and compared observed and permuted residual PR on the same chemically de-redundant support.

The row-norm-preserving null started from the same observed residual support. Independent Gaussian vectors were projected into the $P - 1$ dimensional target-zero-sum subspace and each simulated row was rescaled to the Euclidean norm of its corresponding observed row. This preserves every ligand-specific residual magnitude and the zero-row-sum constraint exactly before column standardization, while removing target alignment and target-specific marginals. We generated 500 realizations for each large matrix.

For residual characterization, targets were ordered by average-linkage clustering of $1 - r_{jk}$. We compared mean absolute within-family and between-family correlations and permuted the fixed broad family labels 5,000 times. The three leading residual target modes were obtained from the same correlation matrix. Their ligand scores were correlated by Spearman’s coefficient with molecular weight, heavy-atom count, cLogP, topological polar surface area, rotatable-bond count and ring count. These descriptive analyses used fixed seeded supports of 12,000 and 15,000 ligands and do not test sequence or pocket mechanisms.

5.4 Physicochemical-axis analyses

The full-panel and matched-support residualizations used seven RDKit descriptors: heavy-atom count, molecular weight, Labute accessible surface area, topological polar surface area, Crippen cLogP, rotatable-bond count and ring count. Descriptors were standardized. For the full anti-target panel, a separate ordinary least-squares model was fit for each target and residuals were assembled before spectral analysis. The nonlinear sensitivity used five-fold shuffled cross-fitting (seed 0) with one histogram gradient-boosting regressor per target (200 iterations, learning rate

0.05, maximum depth 4); no ligand’s fitted value came from a model trained on that ligand. The same protocol was repeated on the matched 74-by-6 support. The narrow-size sensitivity retained molecules with 17–22 heavy atoms, and ligand efficiency divided each score by heavy-atom count. A separate cached diagnostic of the shared score direction used molecular weight, cLogP, topological polar surface area, hydrogen-bond donor and acceptor counts, rotatable-bond count and molecular volume. Pocket volumes were read from the frozen 44-pocket structure summary; 36 targets had complete volumes. The reported univariate statistic correlates pocket volume with the absolute correlation between each target score and the per-ligand panel mean, not with a principal-component loading. The 13-descriptor pocket model was evaluated by leave-one-target-out prediction.

5.5 Preprocessing and experimental-noise sensitivities

For the antitarget panel we repeated the column-standardized analysis with target-median imputation ($r_{\text{PR}} = 1.835$), complete cases (9,297 ligands; 1.320), pairwise-complete Pearson correlations (1.788) and pairwise Spearman correlations (1.686). The complete-case estimate changes the chemical support and is not a like-for-like imputation test. We therefore compared complete and excluded ligands using six RDKit descriptors, standardized mean differences and Butina-cluster coverage, and tabulated missing cells per row and target. Censoring sensitivities treated source-clipped zero cells as missing, removed every affected ligand or removed the most affected target decile. For interaction spectra we compared score-scale centering followed by standardization (9.303), standardization followed by centering (9.378), and robust median/interquartile-range scaling followed by centering (9.462). These analyses test analysis-pipeline sensitivity rather than define alternative primary endpoints.

For the primary 74-by-6 ChEMBL block, target-median imputation was supplemented by 100 posterior multiple imputations from chained Bayesian-ridge regressions. Imputed values were bounded by the observed range of each target. We recomputed both raw and residual PR dimensions and all downstream preference metrics in every completed matrix. Central 95% ranges across imputations describe missing-value sensitivity and are not interpreted as frequentist confidence intervals.

Different-document ChEMBL repeats for the same compound–target pair provided a conservative repeatability scale. From 279,012 paired records, the median absolute difference was 0.47 pChEMBL and the implied single-measurement Gaussian standard deviation was 0.493. For each

of the six matched targets, this standard deviation was divided by that target's experimental standard deviation. Independent Gaussian noise at the resulting target-specific scales was added to the standardized docking block in 2,000 simulations before recomputing PR dimension. This calibration includes inter-laboratory and condition variation; it is not a pure technical replicate variance or a complete experimental noise model.

5.6 Chemical- and target-composition sensitivities

Target-subsampling curves used 200 subsets without replacement at each reported size on one seeded 12,000-ligand support; the complete-panel points used all eligible ligands. Uniform sampling was paired with broad family-stratified sampling based on the versioned data/target_families.csv manifest. Stratified allocations were proportional to family size and included at least one target from every family whenever the requested subset size permitted. Both the column-standardized and two-way-centered residual PR dimensions were recomputed after every draw.

Leave-one-target-out analysis used all 12,651 Docking-44 ligands and one seeded 15,000-row DOCKSTRING support; complete-panel estimates on the same support were plotted as references. The resulting minima, medians and maxima are composition sensitivities, not confidence intervals for a target superpopulation. Ligand-subsampling curves used 50 subsets. For the DOCKSTRING chemical-cluster sensitivity, Bemis–Murcko scaffolds were generated with RDKit; acyclic molecules were treated as singletons. One hundred cluster-bootstrap replicates were compared with 100 molecule-bootstrap replicates on each of five independent 15,000-molecule supports. The first support used the prespecified seed 0; the remaining support seeds were drawn reproducibly from that generator.

5.7 Expanded matched target-preference benchmark

The operational endpoint used the 691 observed exact-relation median ChEMBL cells in the 137-by-38 sparse overlap. For each ligand and every pair of observed targets, a prediction was correct when the more negative docking representation corresponded to the larger experimental pChEMBL value. Exact pChEMBL ties were excluded, leaving 2,522 comparisons. The statistic was the mean of per-ligand pairwise accuracies, giving each ligand equal weight; a predicted score tie received half credit and pair-weighted accuracy was secondary. The broad all-endpoint and human binding Ki/Kd estimands were designated together during revision and are reported

without selecting between them by outcome; this designation was not preregistered. The Vina energy sign convention—more negative predicts stronger binding—was fixed before examining the results.

Nine operational representations decomposed offsets, scales and priors. Absolute Vina used the original scores. For docking score X_{ij} , the external training fit supplied target mean m_j , ordinary target standard deviation s_j^X , residual target standard deviation s_j^R , grand mean g and each evaluation ligand’s row mean r_i over all 44 docking targets. We evaluated unscaled target-centered $X_{ij} - m_j$, unscaled two-way-centered $X_{ij} - m_j - r_i + g$, column-standardized $(X_{ij} - m_j)/s_j^X$, target-centered residual-scaled $(X_{ij} - m_j)/s_j^R$, and fully standardized residual $(X_{ij} - m_j - r_i + g)/s_j^R$ scores. The first two must give identical within-ligand target orders by Eq. 4; the middle comparison isolates the residual-scale choice, and the last adds the ligand-offset by inverse-scale tilt. Missing evaluation scores were filled by target means from the same external fit. Thus, every docking transformation and imputation parameter was fitted without any of the 137 evaluation ligands.

A docking target-prior baseline assigned every evaluation ligand the target means estimated from the 12,514 external Docking-44 ligands. An external experimental prior ranked target mean pChEMBL values estimated from the 96,744 exact-relation ChEMBL ligands outside the evaluation cohort. A final, deliberately stringent diagnostic prior used target mean pChEMBL within the broadly profiled evaluation cohort, fitted with the evaluation ligand’s entire Bemis–Murcko scaffold cluster held out. This is a supervised observation-process diagnostic, not a deployable target-fishing rule.

Uncertainty was estimated by 5,000 ligand bootstraps, 5,000 Bemis–Murcko scaffold-cluster bootstraps over 95 matched clusters and 5,000 fingerprint-based Butina-cluster bootstraps over 121 matched clusters. Paired comparisons resampled per-ligand accuracy differences. Conservative cluster intervals were unions of the Murcko and Butina intervals. A post hoc equivalence sensitivity asked whether the conservative 90% interval lay strictly inside symmetric ± 0.02 or ± 0.05 accuracy margins; neither margin was prospectively registered. The normal-approximation minimum detectable difference used two-sided $\alpha = 0.05$ and 80% power. For absolute, column-standardized and residual scores, 5,000 ligand-identity shuffles broke molecule-specific score–experiment correspondence while preserving target columns; 5,000 within-ligand outcome permutations supplied a second null. One-sided probabilities used the finite-permutation correction. A chemical-dependence sensitivity generated 25 supports by selecting one ligand

per Bemis–Murcko cluster and ran 200 identity permutations within every support. Leave-one-target-out analysis recomputed paired representation differences after deleting each observed target; its range and jackknife approximation describe fixed-panel composition sensitivity, not inference to a random target population. Sensitivities used minimum absolute experimental differences of 0.10, 0.50 and 1.00 pChEMBL, minimum coverage of three, five or six observed targets, leave-one-target-out deletion, human binding assays, human binding Ki/Kd, pair weighting and removal of cells whose repeated measurements spanned more than one pChEMBL unit. A fixed-support same-endpoint sensitivity kept the 137 ligands, 38 targets and docking transformations unchanged, computed comparisons separately within Ki, Kd, IC50 and EC50, averaged endpoint-specific accuracies equally within ligand, and then averaged ligands equally. It therefore avoids comparing unlike endpoint types but can count the same target pair in more than one endpoint stratum. We therefore added a duplicate- collapsed estimator that first averaged correctness for each unique ligand and unordered target pair across eligible endpoint strata, then averaged pairs within ligand and ligands equally. Coverage-stratum intervals used the same cluster bootstrap and are pointwise; the nested strata are not independent. The separate dense 74-by-6 analysis retained its 100 multiple-imputation sensitivity for spectral quantities and is reported in Supplementary Table S3.

5.8 Software and reproducibility

The new analyses and figures were run with Python 3.12.11, NumPy 1.26.4, pandas 2.3.2, SciPy 1.16.2, scikit-learn 1.7.2, Matplotlib 3.10.6 and RDKit 2025.03.6. The release’s `analysis/build_evidence.py` regenerates the centering ladder, preprocessing controls, experimental sensitivities, the expanded observed-pair benchmark, subsampling results and DTI associations in `results/evidence_summary.json`; it also embeds explicitly labelled summaries from the frozen upstream docking/rescoring stages. `analysis/make_figures.py` regenerates all main and supplementary figures. The generic `analysis/spectral_audit.py` command-line tool applies the released audit to a user-supplied dense CSV or TSV score matrix. The package also exports the operational representations, assay sensitivities and audit protocol as machine-readable CSV tables. The core release reproduces the reported analysis from frozen matrices and summaries; it does not rerun receptor preparation, docking or RF-Score pose rescoring. A checksum manifest, pinned requirements, environment file, Dockerfile and Makefile provide input verification and one-command regeneration. Random seed 0 initializes all stochastic procedures; derived support and

694 replicate seeds are stored in the evidence ledger.

695 OpenAI Codex and Anthropic Claude were used during manuscript preparation for code
696 review, literature-search assistance and language editing. They were not treated as sources of
697 scientific evidence. The authors inspected the underlying data and code, verified the reported
698 claims and citations, and retain responsibility for the work.

699 **List of abbreviations**

700 DTI, drug–target interaction; KNN, k-nearest neighbors; PC, principal component; RMT, random
701 matrix theory; PR, participation ratio.

702 **Declarations**

703 **Ethics approval and consent to participate**

704 Not applicable.

705 **Consent for publication**

706 Not applicable.

707 **Availability of data and materials**

708 The source antitarget dataset is available with Nikitin et al.[10]; DOCKSTRING scores and poses
709 are available from the DOCKSTRING release.[9] DAVIS and ChEMBL are publicly available from
710 their respective source repositories. The derived matrices, frozen summaries, source manifest
711 and analysis code are available without registration under an OSI-approved code license in
712 the versioned GitHub repository and Zenodo archive.[29, 30] The exact submission package
713 corresponds to GitHub tag v1.4.0. Upstream datasets retain their stated licenses and are
714 identified by source URL, version and checksum in the manifest.

715 **Competing interests**

716 The authors declare that they have no competing interests.

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Authors' contributions

A.L. conceived the study, performed the analyses and drafted the manuscript. V.S. and M.F. contributed to study design and interpretation. All authors reviewed the manuscript and are responsible for the final submitted version.

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